

OVERCOMING THE PROBLEM OF E-WASTES

Introduction:

As India is moving towards the growth in electronics industry with its production value being estimated over 8.2 trillion Indian rupees which is an increase of nearly 2 trillion from the previous year. India generates a significant amount of e-waste, with estimates ranging from 5 to 17 lakh tonnes annually [5]. This problem is not only associated with India but also with the globe. The United Nations projects that global e-waste generation will reach 82 million metric tons by 2030. This represents a significant increase from the 62 million metric tons generated in 2022. The problem of managing and disposing the e-wastes arises with every innovation in the field of electronics. The safety waste disposal techniques must be followed. This report is aimed to showcase the problems by these wastes and discuss few of the measures which can be followed to reduce the problem caused by E-wastes. More policy oriented approach is appreciated than just saying 'rules'.

E-Wastes and the reason for their increased concerns:

Electronic waste (e-waste) is any broken or unwanted electrical or electronic appliance, including computers, consumer electronics, phones, medical equipment, toys, and other items discarded by the original users or have reached their "end of life" [6]. The E-wastes are the raising concerns because it pollutes the environment and human health. E-Wastes consists of the toxic chemicals such as lead, mercury, cadmium, chromium, arsenic and many more. Burning these e-wastes along with normal wastes causes the humans to inhale toxic gases of above chemicals. Lead and mercury affect the brain development while cadmium and chromium play a key role in the damaging of kidney [1]. When e-waste is dumped into rivers or near water sources, toxic substances mentioned above dissolves in water. This contaminates drinking water and harms aquatic life by disrupting their biological processes. Hazardous chemicals from e-waste seep into the soil, reducing its fertility. This affects crop quality and growth, ultimately impacting food safety and agricultural productivity. The e-wastes mainly comes from the mobile phones production. The survey [6] says that 51.6 percent people doesn't know what to do and have just stockpiled their devices. These are the problems caused by e-wastes which affects the Earth.

Proposed methods:

Some of the methods to reduce e-wastes include many paper publications. Paper [3] suggests a waste basket enabled with smart mechanism only to accept the electronic wastes. It uses ultrasonic sensors and an ESP8266 module. When an object is recognized, the bin allows the cap to open when the garbage reaches a set threshold, information flows to a Raspberry Pi module. The house numbers where trash must be collected is then displayed on the monitor. The main objectives of the project are to lessen the need for human labour, encourage cleanliness and improve the environment and also classifies the e-wastes from ordinary wastes if people are unaware of it. Android application that uses You Only Look Once

version 7 to automatically detect the type of trash is the main goal of this research (YOLOv7) [1] list out the components of identified trash, and provide the classification of components as recyclable and non-recyclable things which will be shown to the administrator dashboard. Finally, the user will get the credits through their e-wallet based on the trash value. If this happens at an international level then people will care more about their disposal methods while earning rewards through apps which use convolutional neural network (R-CNN) to detect the size and type of the waste through images[2] a public-private partnership model and blockchain-based tracking of e-waste creation for effective disposal. It utilizes Blockchain mechanism for creating an immutable tracking system with security and transparency for e-waste. This system ensures accountability and traceability throughout the e-waste management process[4]. These are technological innovations but would only be effective if people cooperate more.

Integrating the proposed methods as policies:

Policy driven approach will be the best for e-waste recycling. Even though the first step has been taken they are not being followed properly. The E-waste (Management) Rules 2016 and E-waste (Management) Amendment Rules, 2018 makes it mandatory for producers and manufacturers to take responsibility for the end-of-life management of product(i.e., The companies are responsible for collecting and recycling some ratio of e-wastes to the total electronic production of them of the year). If more stricter and framed guidelines are provided, people will look more into engaging themselves in separating e-wastes from ordinary wastes. Imposing fines when they don't follow the rules must be an important part of policy framework.

The responsibility of youth and engineers:

Young people and engineers play a crucial role in managing e-waste responsibly. They can develop innovative and affordable recycling solutions that make e-waste management more efficient and accessible, especially in underserved areas. By conducting hands-on workshops and repair camps, they can encourage the habit of reusing and repairing old electronics instead of discarding them. Students can also participate in research projects focused on developing eco-friendly materials and sustainable design practices to reduce future e-waste generation. Additionally, by collaborating with local communities, they can organize e-waste drives, awareness campaigns, and interactive sessions to educate people about the environmental impact of improper e-waste disposal. These actions, supported by eco-clubs and academic institutions, can drive meaningful change toward a more sustainable future.

Conclusion:

The growing challenge of e-waste is an alarming issue that demands urgent and coordinated action at multiple levels. With the rapid advancement of technology and increased dependence on electronic devices, the volume of e-waste generated is rising at an unprecedented rate. This not only threatens environmental health but also poses serious risks to human well-being due to the presence of hazardous substances. While various

technological innovations, such as smart bins, AI-based classification systems, and blockchain tracking, offer efficient solutions for e-waste management, they can only be effective when supported by public awareness and proper implementation. A strong policy-driven approach, with strict guidelines and penalties for non-compliance, is essential to enforce responsible behaviour among producers and consumers alike. Furthermore, empowering youth and engineers to take initiative through eco-clubs, research, and community outreach can create a culture of sustainability. By integrating innovation, policy, and active participation, we can significantly reduce the harmful impacts of e-waste and build a cleaner, healthier, and more environmentally responsible society.

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